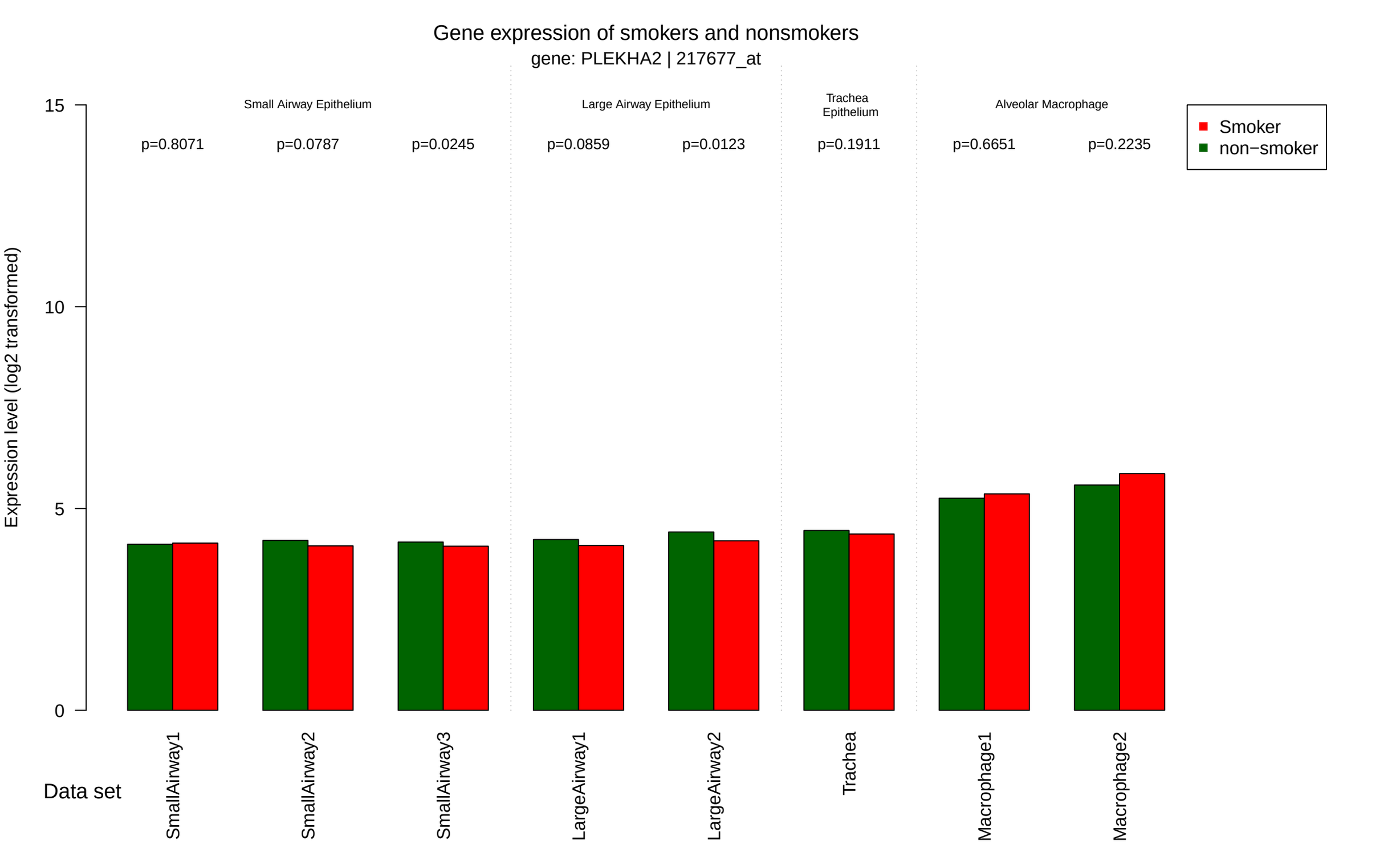
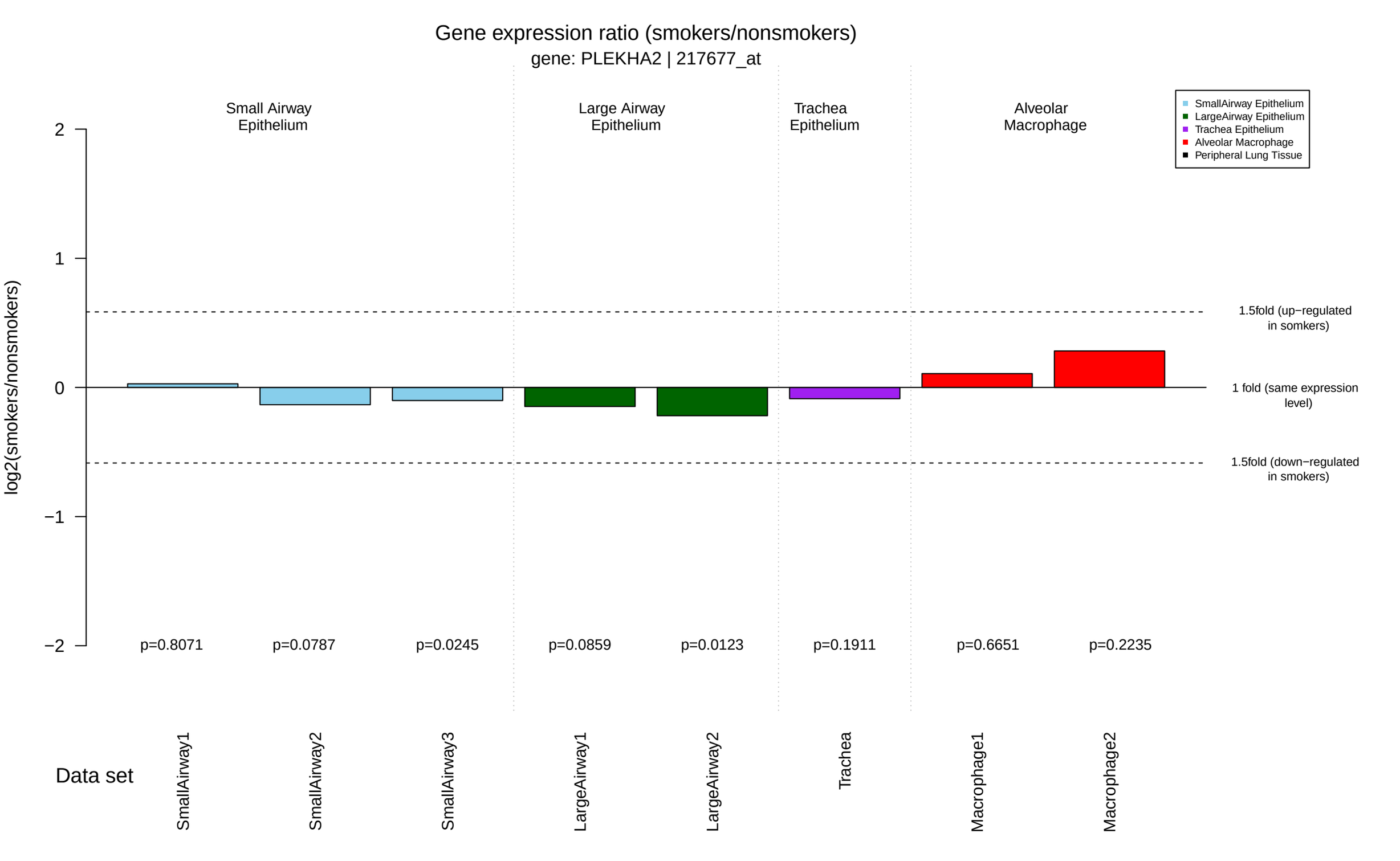
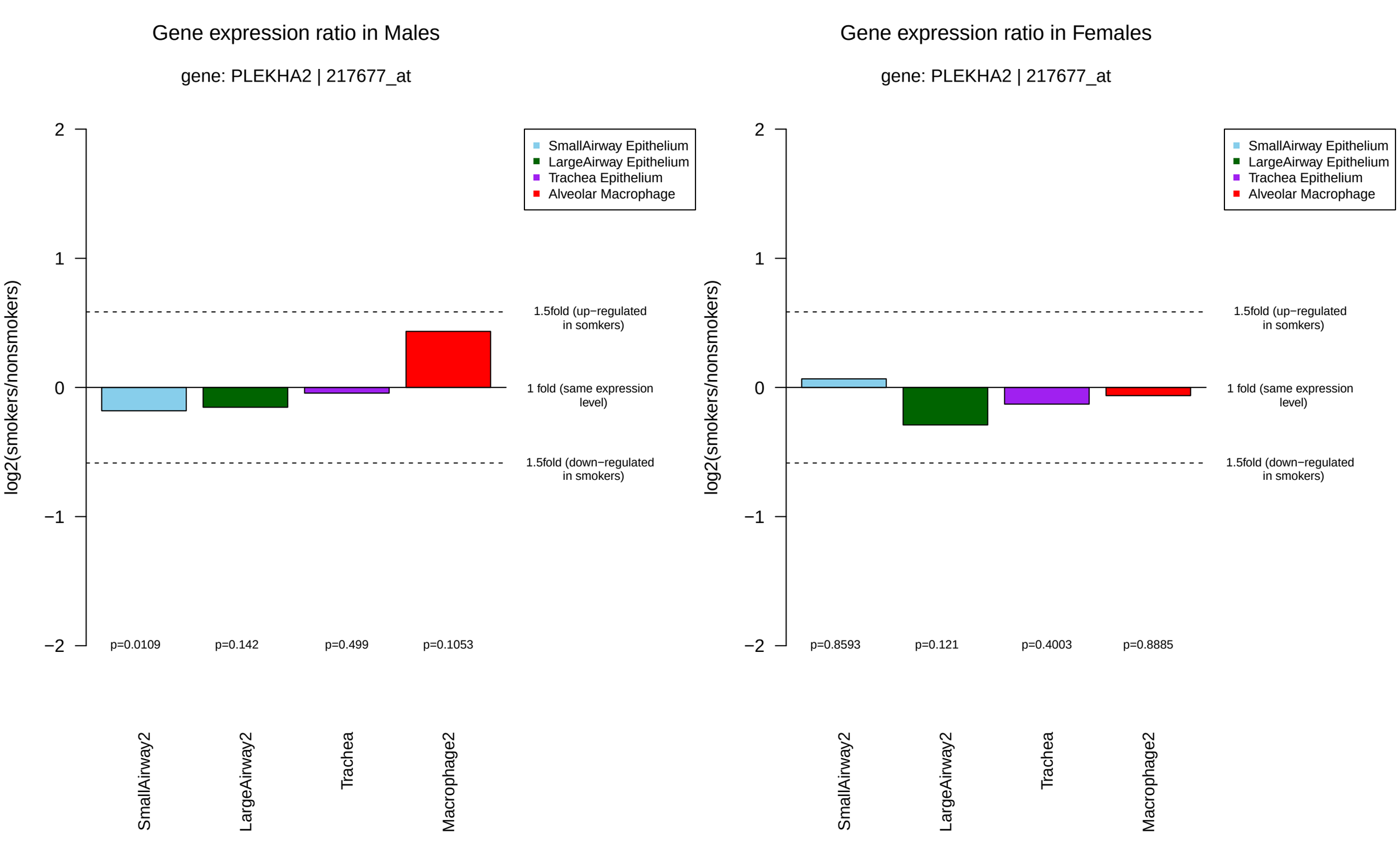
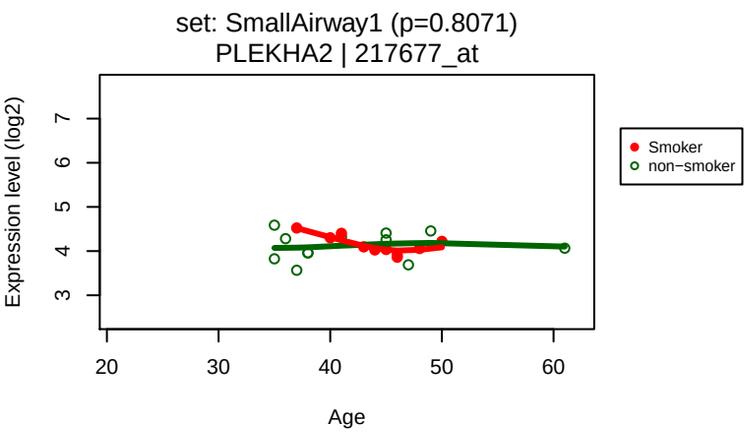
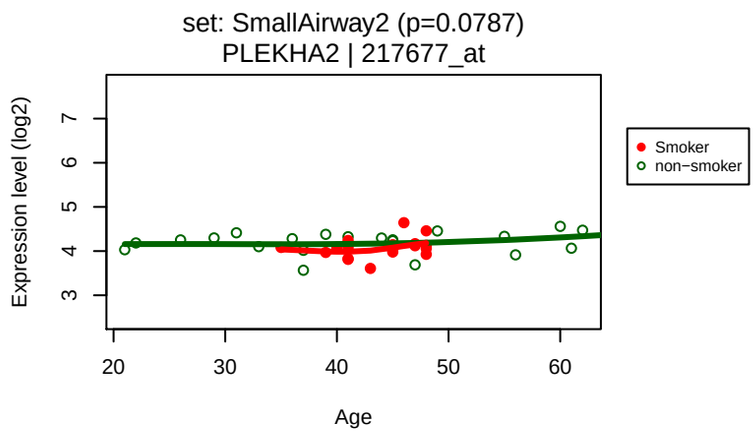
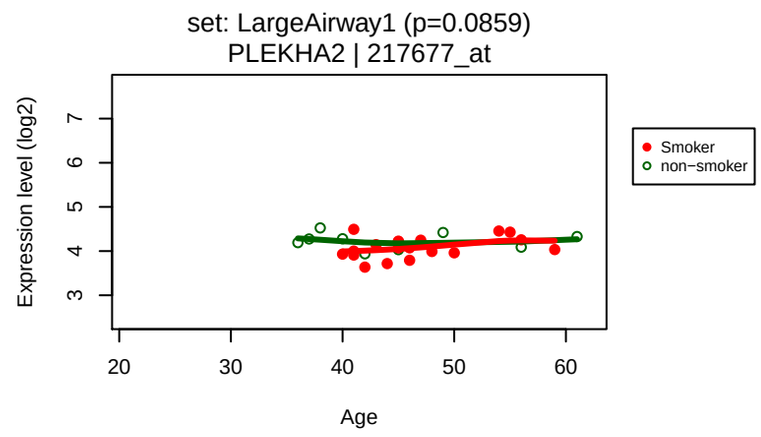

Gene expression of smokers and nonsmokers
gene: PLEKHA2 | 217677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8071
p=0.0787
p=0.0245
p=0.0859
p=0.0123
p=0.1911
p=0.6651
p=0.2235
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

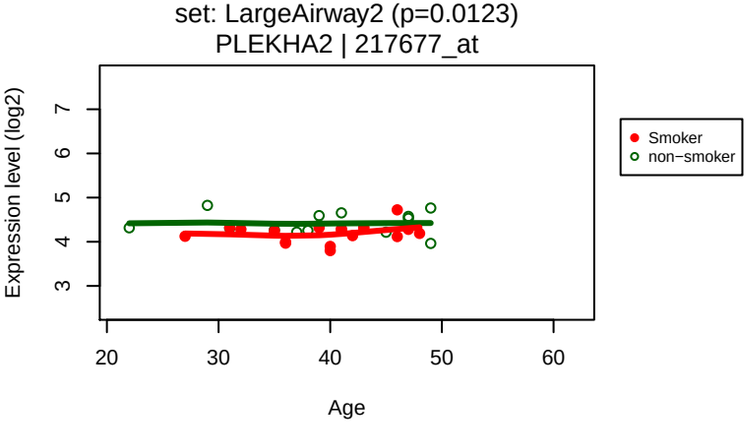

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA2 | 217677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8071
p=0.0787
p=0.0245
p=0.0859
p=0.0123
p=0.1911
p=0.6651
p=0.2235
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

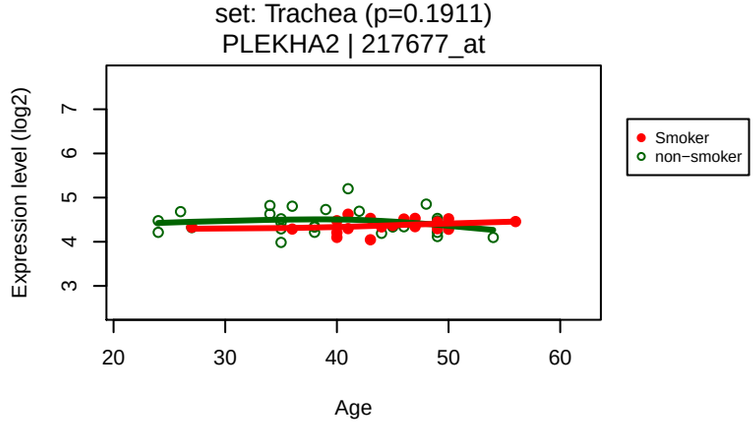

Gene expression ratio in Males
gene: PLEKHA2 | 217677_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0109
p=0.142
p=0.499
p=0.1053
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA2 | 217677_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8593
p=0.121
p=0.4003
p=0.8885
SmallAirway2
LargeAirway2
Trachea
Macrophage2

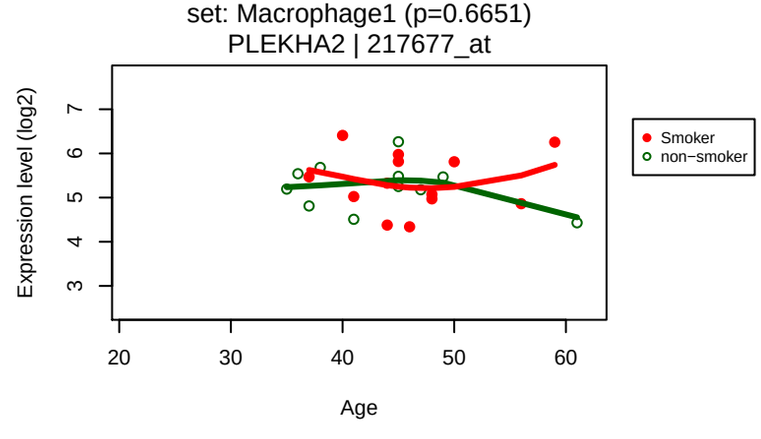

set: SmallAirway1 (p=0.8071)
PLEKHA2 | 217677_at
Expression level (log2)
Age
Smoker
non−smoker

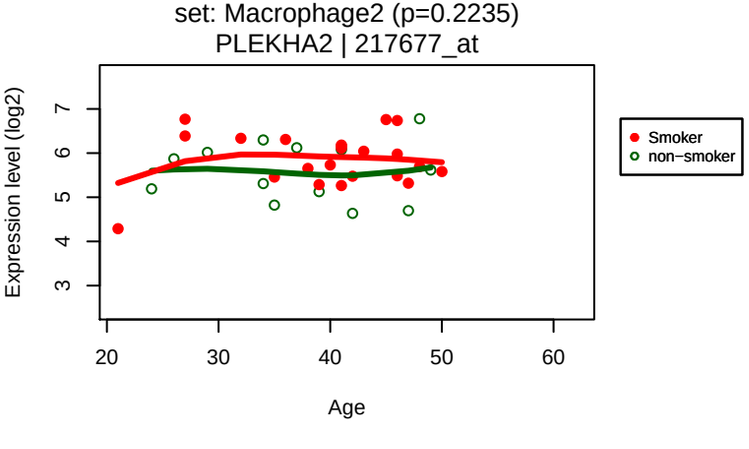

set: SmallAirway2 (p=0.0787)
PLEKHA2 | 217677_at
Expression level (log2)
Age
Smoker
non−smoker

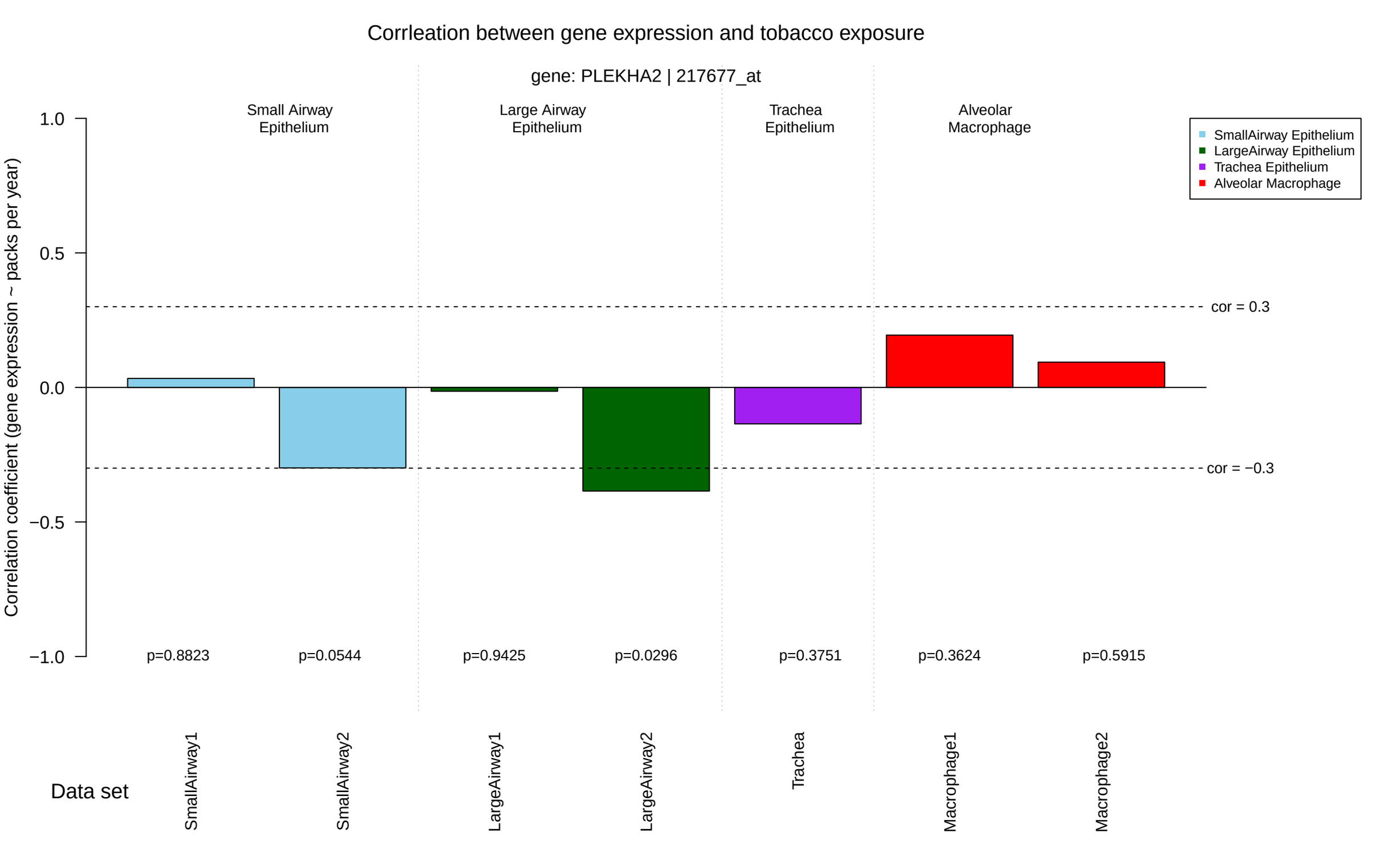

set: LargeAirway1 (p=0.0859)
PLEKHA2 | 217677_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0123)
PLEKHA2 | 217677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1911)
PLEKHA2 | 217677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6651)
PLEKHA2 | 217677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2235)
PLEKHA2 | 217677_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA2 | 217677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8823
p=0.0544
p=0.9425
p=0.0296
p=0.3751
p=0.3624
p=0.5915
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2